

Automated Painting Rover (APR)

Research Findings with Sources — Market, Labour Cost, and Technology Stack

Ref. No.: INC25ETS084848 | **Prepared:** 31 August 2026 | **Scope:** Q1 Existing products · Q2 Telangana labour cost · Q3 Technology stack

Note on sourcing: Every factual claim below is followed by its source. Full URLs are listed in Section 4 so any figure can be independently verified.

Q1 — Is any automated painting rover/device available to buy today?

Short answer: Yes — automated painting robots are real, commercially sold products in 2026. But **none can be bought on Amazon, Flipkart or any consumer marketplace**. They are sold business-to-business, by direct enquiry, through manufacturers and industrial portals such as IndiaMART, TradeIndia and Alibaba. Critically, **there is already at least one Indian company selling exterior façade painting robots** — so the team must not claim this space is empty.

1.1 Consumer marketplaces (Amazon / Flipkart) — nothing available

Searches on consumer e-commerce for "wall painting robot" return only decorative canvas prints, posters, children's toys, and the Scribit wall-drawing plotter (a pen device that draws on walls, not a painting machine). No automated wall-painting rover is listed for retail sale.

Verified by search of Amazon listings; results returned wall-art canvases and the Scribit drawing robot only. Sources [1][2].

What is purchasable in India at retail-ish prices are **airless spray painting machines** — powered sprayers, not robots. They still need a human to hold and aim them:

Product (IndiaMART listings)	Price
Vision Airless Painting Machine, semi-automatic, 2 LPM	₹25,000
Dhruv 409EC Wall Painting Machine, 2.1 LPM, airless	₹24,000
Wall Painting Machine (S S Coating, Pune)	₹1,50,000
Wall Painting Machine (National Enterprises, Faridabad)	₹2,50,000
Blastclean Wooden Robotic Painting System (BCS ROB 01)	₹1,21,000
200L Automatic Barrel Painting System, airless, mild steel	₹7,50,000

Source: IndiaMART product category listings for Wall Painting Machine and Automatic Painting Machine [3][4].

1.2 Indian companies already building painting robots

This is the most important part of the research for the team. Four Indian players are active:

(a) Vividobots — Chennai — the closest direct competitor

Vividobots Pvt Ltd, incorporated 2021, builds façade robots for high-rise buildings. Two products:

- **Wiperque** — a façade cleaning robot, already in the market. Handles flat exterior surfaces on buildings up to 30 floors, and is built to tolerate wind speeds around 55 km/h. The company describes its launch at the Tamil Nadu Global Startup Summit as the debut of India's first automated façade painting and cleaning robot.
- **Vibrantque / Vibrantque-XT** — described by the company as India's first autonomous exterior painting machine for high-rise buildings. It carries a 40-litre onboard paint canister, supports multi-colour painting in one session, and has a robotic arm able to reach projections and window edges. As of a February 2026 interview it was in testing with pilot demonstrations imminent. It is stated to work with oil-based, water-based and acrylic paints.

Funding and validation: ₹1.85 crore raised (Startup Singam Season 2); a seed round reported at ₹1.47 crore from Inflection Point Ventures, Delhi; ₹15 lakh from the Startup India Seed Fund Scheme via Sathyabama TBI; ₹10 lakh

debt from NIDHI-DST through Cultiv8 Coimbatore; grants from EDII-TN. Incubated at Anna Incubation Centre and MCC-MRF Innovation Park; graduated from Brigade REAP 2023 cohort; selected in the top 10 of the Mahindra LEAP programme.

Sources: Vividobots company website [5][6]; Indian Startup Times interview, Feb 2026 [7]; Way2World profile [8].

Be prepared for this question: "An Indian company already sells an exterior high-rise painting robot — how is yours different?" A credible answer must be specific. Vibrantque is a **suspended/gondola-style** system working from the roof down with a large onboard canister; APR is a **self-adhering wall-climbing rover** using vacuum adhesion. Those are genuinely different mechanical approaches with different cost, setup and access profiles — but the team must say so plainly rather than claiming the space is empty.

(b) Pace Robotics — Bengaluru

Builds the **Centa Painter**, a modular construction robot performing putty application, sanding and painting on interior walls and ceilings. The company claims roughly 10× the speed of traditional methods with about 80% less manpower, and a 3–5× reduction in cost of work. Technically it is positioned one metre from a wall edge by joystick, then aligns itself and works autonomously; for ceilings it is fully autonomous. It carries in excess of twenty sensors and uses LiDAR plus camera feed with an NVIDIA processor to build a local map and localise itself.

Sources: Pace Robotics website [9]; CircuitDigest India Tech Tour interview, May 2026 [10].

(c) Endless Robotics — Bengaluru (with Hyderabad projects)

Built **Walt**, an interior painting robot the company claimed paints up to 30× faster than a human painter. Notably for this project, Endless Robotics **targeted mid-level residential projects in Hyderabad** and reported completed projects with Meenakshi Group and Kalpataru Group. The company estimated the total addressable Indian painting market at about USD 3.6 billion, roughly a third of it in major cities, growing around 25% CAGR. Walt scans the interior, reconstructs a 3D model, and is programmed to skip non-paintable areas such as windows, switchboards, beams and doors. The company stated a medium-term intention to move into exterior painting, glass cleaning and high-rise maintenance.

Source: theBUIDL article on Endless Robotics [11]. Note: this report dates from 2019, so current company status should be re-verified before citing it as live competition.

(d) Eternal Robotics — MYRO

An Indian-origin company producing MYRO, an indoor painting robot that takes commands from a tablet and executes a work order from a floor plan; not fully autonomous.

Source: product write-up [12].

1.3 International commercial products

Company / product	Country	What it does and key verified figures
Okibo (EG7)	Israel	Fully autonomous interior painting and drywall finishing. Uses patented AI-guided 3D scanning and real-time modelling; needs no BIM, marking or measurement prep. Finishes an average of about 1,000 sq ft per hour; battery-operated and untethered. Company reports over one million square feet covered across Europe. Raised \$7.85M in March 2024; investors include Pi Labs and Saint-Gobain. An investor writeup states one robot can cost 50% less than human labour and states 350 m²/hour capability.
Legend Robot	China	Latex paint spraying robots up to 6.2 m reach for interior high-rise work; claims over 95% painting coverage and daily output of 2,400–3,000 m².
PaintJet	USA	Robotic spray system mounted on a lift for building and warehouse exteriors — automates the spraying, not the access.
Robosurf	Italy	Autonomous mobile robots and robot painters for construction, industrial and naval sectors; modular, no-code interface. Founded 2018.
Atombrush	Singapore	Robotics for automating building painting, addressing worker shortages.

Sources: Okibo website and Shadow Ventures investment note [13][14]; Legend Robot product pages [15][16]; Standard Bots market guide [17]; CB Insights company profiles [18][19].

1.4 Wall-climbing robots — APR's exact category

This is the category APR belongs to, and it is worth knowing precisely, because the panel may ask whether vacuum climbing is proven.

HausBots (United Kingdom) — the key reference point

A University of Warwick spin-out. Their robots use a proprietary, patent-pending adhesion system called **AEROGRIP**, which creates a pressure differential using airflow rather than a sealed suction cup. Because it does not need to maintain a seal, it can climb rough surfaces and cross obstacles. Verified specifications:

Model	Verified figures
HB1 (earlier model)	Four rubber-tyred wheels; tethered to a ground control unit by a 110 V electrical tether; payload up to 6 kg; generates about 42 kg of force into the surface (≈ 412 N); 30 m range from the ground, potentially unlimited if tethered from the roof. Attachments include brush, robot arm, airless spray and concrete surveying equipment. Two integrated fans draw air from beneath the front and rear and expel it rearward to create downforce.
HB2 / HB3 (current)	Powered by patented AeroGrip. HB2 generates around 350 N of force. Works on both magnetic and non-magnetic surfaces including stainless steel, concrete, brick and composites (FRP). Tethered from the floor for power, so it can run continually.

Sources: HausBots website [20][21]; New Atlas [22]; Pharmaceutical-Technology / GlobalData report [23]; company LinkedIn stating use for exterior painting, warehouse painting and inspection [24].

Why this matters for APR's numbers. HausBots' HB2 — a commercially deployed wall-climbing robot — generates about **350 N** of adhesion. APR's requirement document targets adhesion of at least 2× rover weight, and the earlier working figure was ≥ 500 N for a 12 kg rover. The HausBots figure is independent, real-world evidence that adhesion forces in the 350–450 N range are physically achievable with airflow-based systems. Citing this makes APR's target look researched rather than invented.

Historical Japanese systems (proof the concept is decades old)

The **Kumagai-Gumi** wall-climbing robot performed shot blasting and painting on vertical surfaces, holding itself to the wall using force generated from vacuum suction, eliminating the need for cables or trolley rails — although safety cables were still used to prevent falls. It covered three to five metres per minute. **Shimizu OSR-I** painted balustrades on high-rise residential and commercial buildings. The **Taisei** painting robot painted vertical structures with automatic detection and avoidance of window panels.

Source: RoboticsBiz survey of exterior wall finishing robots [25].

Academic work in India also exists: VIT University (Chennai) published development of an exterior wall painting robot, PAINTBOT-EW, using a robot plus suspension system.

Source: Indian Journal of Science and Technology, Vol 9(38), 2016 [26].

1.5 Price landscape (for the "what would this cost" question)

Segment	Reported price range
Basic wall painting robots (manual control, simple spray)	\$3,000 – \$7,000
Semi-automatic (motorised, partial automation)	\$8,000 – \$15,000
Fully automated (AI-assisted, advanced sensors)	\$16,000 – \$40,000+
Typical business "sweet spot" per one market guide	\$30,000 – \$100,000
Overall painting-robot range in 2026	\$5,000 – \$200,000+
Wall plastering/coating robots on Alibaba (bulk)	\$1,180 – \$38,000

Sources: Standard Bots 2026 pricing guide [17]; Painting Medford Oregon price breakdown [27]; Alibaba showroom listing [28]. These are vendor and market-guide figures, not audited prices — treat as indicative.

Q2 — Labour cost to paint an 8 ft × 8 ft wall in Telangana today

Wall area = 8 ft × 8 ft = 64 square feet. Below are three independent methods. They agree closely, which is what makes the answer defensible.

2.1 Method A — Per-square-foot labour-only rates

Published labour-only rates (paint supplied separately) for India and Hyderabad in 2026:

Work type	Labour rate (₹/sq ft)	Source
Distemper	₹4 – ₹7	Solve24, India 2026 [29]
Emulsion	₹8 – ₹14	Solve24, India 2026 [29]
Putty + base coat	₹6 – ₹10	Solve24, India 2026 [29]
Texture	₹18 – ₹45	Solve24, India 2026 [29]
Labour share of all-in rate (Hyderabad)	₹4 – ₹12 (40–70% of rate)	BestPainters, Hyderabad 2026 [30]
Labour-only quoted rate (incl. prep, putty, primer, coats)	₹10 – ₹14	AapkaPainter calculator [31]
Distemper labour, Hyderabad	₹8 – ₹15	Asian Paints / SCIB Hyderabad page [32]

For 64 sq ft — labour only:

Distemper, basic: $64 \times ₹4-7 = ₹256 - ₹448$

Emulsion, standard: $64 \times ₹8-14 = ₹512 - ₹896$

Emulsion + putty & primer prep: $64 \times ₹14-24 = ₹896 - ₹1,536$

Most representative labour-only figure for a repaint: ₹500 – ₹900

2.2 Method B — Daily wage basis

Statutory minimum wages (Telangana, effective 1 June 2026)

The Telangana Government notified a comprehensive minimum wage revision on 30 May 2026, effective 1 June 2026, under a three-zone system. **Hyderabad falls in Zone I.** The ILO noted this as Telangana's first comprehensive minimum wage fixation since state formation, covering more than 9.2 million wage workers.

Skill category	Zone I (monthly)	Daily (÷ 26 days)
Unskilled	₹16,000	₹615
Semi-skilled	₹17,000	₹654
Skilled (<i>a painter falls here</i>)	₹18,500	₹712
Highly skilled	₹20,000	₹769

Sources: Telangana Labour Department notification dated 30 May 2026 as reported [33][34]; ILO statement [35]; zone definitions — Zone I covers Municipal Corporation areas including Hyderabad.

Actual market rates (higher than statutory minimums)

Worker	Market rate per day (India, 2026)
Helper	₹500 – ₹700
Skilled painter	₹800 – ₹1,300
Painting contractor	₹1,500 – ₹2,200+

Source: Solve24 painter salary survey, India 2026 [29]. A comparable Chennai-specific guide gives skilled painters ₹800–₹1,100/day and helpers ₹400–₹550/day [36].

Productivity reasoning: a 64 sq ft wall is a small job. A skilled painter with basic prep and two coats will finish it in roughly 2–3 hours, i.e. about one-third of a working day.

One-third day of a skilled painter at ₹800–₹1,300/day = **₹265 – ₹430**

But — painters do not travel for a two-hour job. In practice a **half-day or full-day minimum** is charged.

Practical daily-wage answer: ₹400 – ₹800 (half day to full day)

2.3 Method C — All-in cost (labour + material), for context

Hyderabad all-in rates in 2026 (labour + putty + primer + two coats):

Category	Rate (₹/sq ft)	Cost for 64 sq ft
Interior, economy emulsion	₹12 – ₹18	₹768 – ₹1,152
Interior, standard/mid-range	₹20 – ₹30	₹1,280 – ₹1,920
Interior, premium (Royale grade)	₹26 – ₹38	₹1,664 – ₹2,432
Exterior (weather-grade paint + scaffolding)	₹13 – ₹50	₹832 – ₹3,200

Sources: AapkaPainter Hyderabad 2026 guide [37]; BestPainters Hyderabad 2026 [30]. Both note Hyderabad rates run roughly 5–10% below Mumbai and Delhi.

Final answer to Q2. To paint an 8 × 8 ft (64 sq ft) wall in Telangana today:

- **Labour only, ground level: roughly ₹500 – ₹900** (₹8–₹14/sq ft emulsion), with a practical floor of ₹400–₹800 because of half-day/full-day minimums.
- **All-in with materials, interior: roughly ₹1,300 – ₹1,900** at standard mid-range quality.
- Statutory basis: a skilled worker in Hyderabad (Zone I) must be paid at least **₹712 per day** from 1 June 2026.

The number that actually matters for APR's business case. A 64 sq ft wall at ground level costs only a few hundred rupees in labour — **a robot cannot compete there, and the team should not pretend otherwise.** The economics change completely with height. Hyderabad sources put **scaffolding alone at ₹1,500 – ₹5,000** for homes with high walls, and exterior rates carry that scaffolding premium. So for the same 64 sq ft placed three floors up, access cost can exceed the painting labour by three to five times, before counting the fall risk. **APR's value is in eliminating access cost and fall risk at height, not in undercutting a ground-floor painter.** Frame the answer that way if the panel asks about cost competitiveness.

Scaffolding figure source: Asian Paints / SCIB Hyderabad painting rates page, which states scaffolding charges typically range ₹1,500–₹5,000 for homes with high walls [32].

Q3 — Technologies APR will use, in detail

APR is not "one AI." It is a stack of purpose-chosen subsystems, each selected because it is the correct tool for that specific job. The pipeline below shows how they connect.

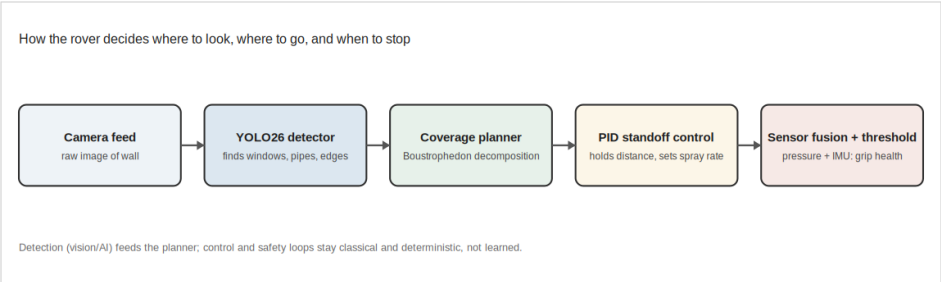


Figure 1. The decision pipeline. Vision/AI feeds the planner; control and safety loops stay classical and deterministic.

3.1 Adhesion — airflow-based vacuum (negative pressure)

Chosen: actively-driven negative pressure using impeller/fan, with a compliant skirt, rather than passive suction cups.

Why: the standard survey of wall-climbing adhesion identifies four primary mechanisms — biomimetic dry adhesion (gecko-style), magnetic adhesion, electrostatic adhesion, and vacuum suction. For concrete, brick and plaster — electrically insulating, non-ferrous, porous and rough — magnetic adhesion is physically impossible and electrostatic is weak. A patent filing states directly that for adhering to structures such as bridge piers, box girders and building exterior walls made of electrically insulating materials, **vacuum adhesion using a vacuum pump or fan is effective among the available methods.**

Why not passive suction cups: HausBots' own technical page states plainly that sealed-suction systems must maintain the seal at all times and are therefore only really effective on smooth surfaces — which is exactly why their AEROGRIP system uses a low-pressure airflow effect instead of a maintained seal, allowing it to work on brick, concrete, glass and metal even when uneven or dusty.

Sources: adhesion mechanism survey [38]; USPTO patent 11408462 on adhesion devices [39]; HausBots wall-climbing robots technical page [20].

Design evidence: HB2 achieves ≈350 N; HB1 ≈42 kg-force (≈412 N) with 6 kg payload. These are real deployed figures supporting APR's force targets.

3.2 Locomotion — tracks over wheels

Chosen: track-based locomotion. Tracks spread load over a larger contact patch, which helps maintain the pressure differential while moving and gives better grip on rough plaster and weld lines. Note that HausBots' HB1 used four rubber-tyred wheels successfully with an airflow system, so wheels are not disqualified — the choice must be justified on surface roughness and seal stability, not asserted.

3.3 Computer vision — YOLO family object detection

Important correction from this research. Earlier project drafts specified **YOLOv8**. That is now two generations old. As of 2026 the current models are **YOLO26** (released 14 January 2026) and **YOLO11** (September 2024, the mature stable release). Ultralytics' own documentation recommends YOLO26 for new projects and lists YOLO11 as the stable production alternative. Presenting YOLOv8 as a 2026 technology choice will look dated to a technical panel.

Recommended: YOLO26, with YOLO11 as the conservative fallback.

Property of YOLO26	Verified detail
Release	January 2026, current Ultralytics generation
Architecture	End-to-end, NMS-free inference — produces final detections directly rather than requiring non-maximum-suppression post-processing

Edge performance	Up to 43% faster CPU inference than YOLO11-N; explicitly engineered for edge and low-power devices
Accuracy span	Nano at 40.9 mAP (1.7 ms on T4) up to Extra Large at 57.5 mAP
Tasks	Detection, segmentation, classification, pose, oriented bounding boxes
Export	ONNX, TensorRT, CoreML, TFLite; INT8/FP16 quantisation

Sources: Ultralytics model documentation [40][41]; Roboflow 2026 model comparison [42]; Viso.ai YOLO lineage analysis [43]; YOLO26 architecture paper [44].

Licensing — worth knowing before commercialisation. YOLO26 ships under **AGPL-3.0**, which requires open-sourcing derivative works, or purchasing a commercial licence for closed-source products. Since APR is intended as a commercial product, this is a real consideration and a good detail to raise proactively — it signals commercial awareness.

Source: Roboflow best object detection models, 2026 [42].

Why not the alternatives

Alternative	Why not chosen
Two-stage detectors (Faster R-CNN)	Higher benchmark accuracy but too slow for real-time inference on a battery-powered embedded board; the lag is unacceptable on a moving rover
Classical edge detection (Canny/contours)	Fragile under the lighting and texture variation of real plaster; no semantic understanding — finds an edge but cannot distinguish a window from a crack
YOLOv12	Attention-centric design gives accuracy gains, but Ultralytics itself notes training instability, elevated memory use and slower CPU throughput, and recommends YOLO11 or YOLO26 for production
Training a detector from scratch	Requires far more labelled data and time than the project timeline allows; fine-tuning a pretrained model on a small custom dataset of wall images is the realistic path

Sources: [42][43]; YOLO edge-deployment survey [45].

3.4 Coverage path planning — Boustrophedon Cellular Decomposition

Chosen: Boustrophedon Cellular Decomposition (BCD), developed by Howie Choset and Philippe Pignon and published at the 1st International Conference on Field and Service Robotics in 1997, with the extended treatment in *Autonomous Robots* (2000).

How it works: the workspace is divided into cells; each cell is covered with simple back-and-forth motions (the name comes from the Greek for the way an ox turns while ploughing). Once every cell is covered, the entire environment is covered — so coverage reduces to finding an exhaustive path through a graph of cell adjacencies.

Why it is the right choice: the authors state the approach is **provably complete**, and validated it experimentally on a mobile robot. A survey of coverage path planning confirms boustrophedon decomposition generates shorter complete coverage paths than trapezoidal decomposition for the same class of environments. A 2024 multi-robot paper confirms it guarantees full coverage in structured environments, minimises unnecessary turns, and has been widely adopted for computational efficiency.

This maps directly onto APR's requirement of at least 90% uniform coverage with no missed or repeated areas — a provable completeness guarantee is exactly what that criterion needs.

Sources: Choset & Pignon, Carnegie Mellon Robotics Institute [46]; Springer chapter [47]; Choset, *Autonomous Robots* 9:247–253 (2000) [48]; Galceran & Carreras survey [49]; multi-robot BCD paper, 2024 [50].

Why not the alternatives

- **Reinforcement-learning exploration** — requires extensive training data or simulation and gives no completeness guarantee. Unacceptable when the deliverable is a fully painted wall, not a mostly-painted one.
- **Spanning-tree coverage (STC)** — a legitimate alternative, but produces more turns and is generally less efficient than BCD on wide, largely obstacle-free surfaces like a building wall.

- **Random or bio-inspired coverage** — less predictable and harder to verify before a live demonstration.

3.5 Motion and spray control — closed-loop PID

Chosen: classical PID control tied to the standoff-distance sensor and traverse speed, holding the nozzle at constant distance so film thickness stays consistent.

Why not a learned control policy: it would need extensive real paint-spraying trial data to train, which the project does not have time to collect. PID is well understood, tunable, computationally cheap in real time, and requires no training phase.

3.6 Safety — deterministic threshold logic, deliberately not AI

Chosen: sensor fusion of vacuum pressure and IMU tilt, with a rule-based threshold trigger that raises an alarm and enters safe-hold the moment a reading crosses a defined limit.

This is a deliberate engineering decision, not a gap — say so confidently. A safety-critical trigger, the one deciding whether the robot is about to fall, must be simple, deterministic and provable before a demonstration. A learned model cannot be argued about the same way; a threshold on a physical sensor reading can be tested and verified. This matches APR's own requirement wording: alarm if adhesion force drops below threshold. Note that even HausBots and the historic Kumagai-Gumi system retained physical safety cables alongside their adhesion systems — redundant mechanical safety is industry-standard practice, not over-caution.

3.7 Onboard compute

Recommended: NVIDIA Jetson Orin Nano. Independent validation: Pace Robotics' Centa Painter uses an NVIDIA processor with LiDAR and camera feed to build a local map and localise. A 2026 robotics guide describes the Orin Nano 8 GB as the default for this class of robotics, running YOLO plus SLAM at usable rates, and notes Jetson is overwhelmingly the platform of choice in 2026 for robots running deep models on-board — roughly 3–10× the cost of a Raspberry Pi 5 but 10–100× the AI capability.

Option	Detail
Jetson Orin Nano Super	\$249 entry point; up to 40 TOPS; runs YOLO + SLAM concurrently
Jetson Orin Nano 2 (announced)	Announced 25 August 2026: 78 TOPS, 8 GB LPDDR5x, 8-core Arm CPU, 2× inference of the Super at the same form factor, 40% less power at matched performance, 40 W TDP. Availability expected first half of 2027 — too late for this prototype, but worth citing as the roadmap.
Software stack	NVIDIA Isaac ROS with optimised ROS 2 packages for localisation, 3D reconstruction and depth estimation

Sources: NVIDIA newsroom press release, 25 Aug 2026 [51][52]; Unite.AI analysis noting H1 2027 availability [53]; RobotForge Jetson guide [54]; NVIDIA developer blog on Isaac ROS [55]; Pace Robotics interview [10].

3.8 Summary technology table

Subsystem	Technology chosen	One-line justification
Adhesion	Airflow-driven negative pressure + compliant skirt	Only viable method on non-ferrous, porous concrete; proven at 350–412 N by HausBots
Locomotion	Track drive	Larger contact patch keeps pressure differential stable while moving
Perception	YOLO26 (fallback YOLO11)	Current generation, NMS-free, edge-optimised, 43% faster on CPU than YOLO11-N
Path planning	Boustrophedon Cellular Decomposition	Provably complete coverage; matches the ≥90% coverage criterion
Standoff/spray control	Closed-loop PID	Deterministic, cheap, no training data needed
Fall detection	Pressure + IMU fusion, threshold rule	Safety logic must be provable, not statistical

Compute	NVIDIA Jetson Orin Nano	Runs YOLO + SLAM on-board; same class Pace Robotics uses
Spray	Airless spray head	Lower overspray, more consistent film thickness
Connectivity	IoT telemetry to operator dashboard	Live status, coverage record, remote stop

4. Source List

All URLs were accessed on 31 August 2026. Figures quoted are as reported by these sources; vendor claims are identified as such in the text above.

1. Amazon search results for wall drawing/painting robots — <https://www.amazon.com/scribit-wall-drawing-robot/s?k=scribit+wall+drawing+robot>
2. Amazon wall-art listings returned for "robot painting" — <https://www.amazon.com/Painting-Canvas-Futuristic-Artwork-Bedroom/dp/B0D6BZ785Q>
3. IndiaMART, Wall Painting Machine category — <https://m.indiamart.com/impcat/wall-painting-machine.html>
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16. Legend Robot, painting robot cost guide — <https://www.legendrobot.com/blog/how-much-does-a-wall-painting-robot-cost-a-practical-buying-guide.html>
17. Standard Bots, wall painting robot price guide 2026 — <https://standardbots.com/blog/buy-automatic-painting-robot>
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21. HausBots, HB2 product page (350 N figure) — <https://hausbots.com/hb2/>
22. New Atlas on the HausBots HB1 — <https://newatlas.com/robotics/wall-climbing-hausbots-hb1-robot/>
23. Pharmaceutical-Technology / GlobalData, HausBots HB1 specifications — <https://www.pharmaceutical-technology.com/research-reports/hausbots-rolls-out-wall-climbing-robot/>
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5. Three Things to Be Careful About

1. **Do not claim nobody is doing this.** Vividobots (Chennai) markets Vibrantque-XT as India's first autonomous exterior high-rise painting machine and has raised institutional funding. Acknowledge them and differentiate on mechanism — self-climbing vacuum rover versus suspended gondola system.
2. **Update YOLOv8 to YOLO26 or YOLO11** in all project documents. YOLOv8 dates from 2023 and two newer generations have shipped since. Also note the AGPL-3.0 licensing implication for a commercial product.
3. **Frame the cost argument around height, not ground level.** Labour for a 64 sq ft ground-floor wall is only ₹500–₹900; no robot beats that. The real cost APR removes is access — scaffolding at ₹1,500–₹5,000 — plus the fall risk. That is a defensible economic argument; "cheaper than a painter" is not.